

The Medium

First principles of the vacuum — why space is a loaded medium and not an empty stage, why a static coupling to it is exactly zero — symmetric or not — and a driven, transient one is not, and why one asymmetric machine gives power, force and weight depending only on where its return path closes.

draft 0.2 · 2026-09-23 (energy reading corrected: \$2, \$3, \$6–\$8, \$10, \$13) · first draft 2026-08-18 · Osika Innovation AB · Alexander Osika, Claude

What this paper is. The front door to the network. Every paper behind it — the gravity law, the field guide, the warp mechanism, the materials, the bench floor, the energy lane — reasons from one ontology: space is a medium, the medium is loaded, matter is a pattern the medium sustains, gravity is what stored energy does to the medium, and the medium can be coupled to only *asymmetrically and in time*. Until now that ontology was stated in each paper's middle and hedged in each paper's opening with "we reason only from known physics." This paper states it first, in one place, with the receipt and register of every link, and names honestly the two things the ontology still owes. Nothing here contradicts a measurement. Several things here are not yet measured, and are tagged.

Every claim carries one of seven tags: **MEASURED** (instrument, published, or receipted in this project) · **DERIVED** (follows from the medium law, receipt named) · **ESTIMATE** (a prior, honestly labelled) · **PRE-REG** (frozen prediction, not yet run) · **OPEN** (the bench decides) · **NULL-STATIC** (predicted zero by the medium law *and* measured zero — a confirmation, not a failure) · **REFUTED** (measured, and contradicts the claim). The split of the last two is new to the network in this paper and is adopted across the shelf: almost every null in the electrogravitics ledger is the first kind, and the old single label read as failure when it was retrodiction.

Space is a loaded medium, not an empty stage. Matter is a standing pattern the medium sustains; gravity is what stored energy does to the medium's rulers and clocks — the weak-field law is derived, factor of two included, and general relativity is its shadow. Because the medium is loaded, a machine can couple to it — but only

asymmetrically and in time: a static coupling is exactly zero, symmetric or not, which is why a century of static capacitors on balances read null, why a diode in equilibrium cannot rectify its own noise, and why a mirror at a nanometre gap changes how fast matter trades with the vacuum but never the level it trades at. Drive the boundary in time — a moving mirror, a flash across an asymmetric capacitor, a handed spiral of coherent energy — and the medium answers: as light (measured: the dynamical Casimir effect, with the driver paying) or as force (every vacuum positive in the electrogravitics lineage is a transient, and the coherent channel has never once been tested). The one measured *static* current in the lineage — 70 W/m^2 at a 33 nm gap, a thousand devices — therefore has an open source, and §6 names the run that finds it. It is one machine with three readings — power out, force out, weight change — depending on what its return path couples to. Building a craft that stands on the medium is therefore not new physics; it is the transient, coherent, consonant version of a device whose static version we already know is null. What is not yet done is the climb — from the incoherent floor at $8\pi G/c^4$ to hover, thirty-seven orders, on one dial: coherence.

PART I — THE MEDIUM

1. Space is a medium

The root paper of this network (The Shape of the Universe) does not describe space as a container with things in it. It describes space as a *constrained network*: equal-reference-length elastic edges meeting three at a time, a foam whose seams are the edges, closing on a boundary screen that is a Goldberg tiling — locally flat and hexagonal, globally closed by exactly twelve pentagonal defects at the vertices of an icosahedron, of order 10^{122} pixels. The one clean number that falls out of the count is the cosmological constant, $\Lambda = 3/N$, with no free parameter. **DERIVED** ($\Lambda = 3/N$) · **OPEN** (which global graph)

The sentence that matters for everything downstream is the one that defines what an edge *is*:

An edge is "not a thing placed *in* space, but part of the frame space is made of ... a local metric element whose length is defined by the energy pattern it carries." — Shape §2

Older physics had a word for exactly this and then lost the right to use it. The nineteenth-century ether was a *wind* — a substance with a rest frame you could measure your motion against — and Michelson–Morley, and every ether-drift bound since, found no wind. The word died with the wind. But the thing physics kept is a medium in the only sense that matters: something with properties at every point, that stores energy, that responds to what is stored in it, and that carries what a body pushes against. Dicke's variable-index vacuum (1957), Volkov's curved-space Maxwell as a dielectric (1971), Sakharov's induced gravity (1967), Puthoff's polarizable-vacuum representation of general relativity (1999–2002), Sonny White's compressible-dispersive vacuum in *Physical Review Research* — all mainstream, all treating the vacuum as a real medium with a real response. The network's word is *the lattice*; the plain word is *the medium*; the honest historical word is *the ether, made precise*: a loaded, elastic, Lorentz-respecting network — not a wind. **MEASURED** (no drift) • **DERIVED** (the network's medium)

That last clause is a constraint the medium *owes*, and this paper says so in §10 rather than hiding it: a medium ontology must be built to respect Lorentz invariance — the null of Michelson–Morley is a constraint, not a mistake (the network's own *ISF/Haramein audit* puts it exactly that way). No paper in the network yet derives boost invariance from the lattice; the scalar law recovers every weak-field ratio of general relativity, which is strong evidence that the covariance is there to be found, and not a proof. Half of the debt has a known answer, recorded in §10: the medium's loading must be $\hbar\omega/2$ per mode — an energy spectrum rising as ω^3 — because that is the only loading that looks the same from every inertial frame. A thermal loading has a wind; the cosmic background's 3.36 mK dipole is exactly one.

2. The medium is loaded

The second thing the root paper says about the vacuum edges is that they are not slack. In the elastic-metric law every vacuum edge sits at a definite reference energy, $E_\star = 2 \cdot m_{\text{edge}} \cdot c^2$ — twice the rest-energy of the edge's own settled content. Matter is settled edge-energy; the vacuum is edges loaded to $E_\star$. The vacuum is therefore a

charged reservoir, not empty space. **DERIVED** (role of $E^\star$ in the law) · **OPEN** (its absolute value — §10)

The mainstream twins of this statement are among the best-measured facts in physics, and the network's energy lane calls them "the measured foundation of our ZPE lane": the Lamb shift; the Casimir force between close conductors (Lamoreaux 1997; Mohideen & Roy 1998); and the dynamical Casimir effect — change a boundary in time and the vacuum yields *real photons* (Wilson et al., *Nature* 479:376, 2011). $\langle E \rangle = 0$ while $\langle E^2 \rangle \neq 0$: the ground state has no particles in it and is not nothing. **MEASURED**

Two things must be said in the same breath, because the network's energy papers say them and the free-energy literature never does. First: the ground state is a *level*, not a stockpile. "The zero-point field cannot be the fuel. It is the ground state; there is nothing beneath it to fall to" (The Optical-Cavity Device §1d). The famous 10^{113} J/m^3 is a bookkeeping error 121 orders of magnitude wide — "anyone quoting the large number as an available resource is quoting the error." What can be exploited is a *step in the level* — "the cavity puts a step in the sea level" (power-cell explainer §5) — but only by moving it: the step is the Casimir energy, released once when the boundary moves ($12 \mu\text{J/m}^2$ at 33 nm) or pumped by a driver (the dynamical Casimir effect). Second — and this corrects the network's own earlier wording — the zero-point floor is not a *cold sink* either. It is the $n = 0$ floor of every mode at every temperature, so it cancels from every net flow of heat: two bodies at one temperature exchange nothing, whatever cavity sits between them. A cavity is a *slow* place, not a cold one (the fluctuation–dissipation theorem, made concrete by Puthoff 1987 — see §3). A heat engine needs a colder *body* — the night sky at 2.7 K is a real one, measured at 50 mW/m² of nighttime generation — and a device that delivers power while running cold must name its cold body. That is the discriminator between a real open-system device and a traced fraud, and it is testable. **DERIVED** (bookkeeping; receipt `puthoff_zpf_balance_run`) · **MEASURED** (sky) · **PRE-REG** (cold-side sign test)

3. Matter is a pattern the medium sustains

If space is a loaded elastic network, matter cannot be a foreign object placed in it. The root paper: matter is "energy that has reached a stable normal form: a persistent pattern that keeps satisfying the local constraints as the network updates." The Electron makes it concrete: the electron is "a single wavelength of light closed on itself in a double loop

— a toroidal, self-confined photon"; leptons and photons are "different states of the same object ... light that has stopped going somewhere and started going around." The charge comes out at e to 0.08% by energy-norm extraction; spin $\frac{1}{2}$ and the 720° return come out of the double loop; the Compton radius comes out as $\lambda_C/4\pi$. **DERIVED** (charge, spin, radius — toy-level, tagged as such in that paper)

What holds the loop closed is the part that connects matter to the loaded medium. Refraction alone fails (the network ran it: `ring_fixed_point_run`, null). What works, at the toy level, is *commitment*: the pattern is a written record on the ledger, and — this is the sentence that ties the energy lane to the matter lane —

"the vacuum's zero-point bath sustains standing patterns; repair dissipates mismatch back; conservation holds on the joint ledger ... The donut stays closed because the pattern is a committed record — the ledger holds it, the vacuum feeds it, and the $\alpha/2\pi$ leakage is the standing cost of being written into a shared world." — The Electron §6

Read plainly: a particle is a pattern the medium continuously *feeds*, and its Coulomb field is the standing leakage of that feed — $\alpha/2\pi$ per port. **OPEN** (conditional; the network's own tag, for the electron's self-structure)

At the scale of the atom this is not a conjecture. Puthoff (*Phys. Rev. D* 35:3266, 1987) showed that the hydrogen ground state is the orbit on which the power the electron radiates equals the power it absorbs from the zero-point field — it lands exactly on the Bohr radius — and quantum electrodynamics says the same thing exactly: in the ground state the vacuum-fluctuation and radiation-reaction contributions cancel, and in an excited state they add, which is spontaneous emission (Dalibard, Dupont-Roc & Cohen-Tannoudji 1982). The flow is large — about 47 nW in and 47 nW out for every hydrogen atom, 3×10^{16} W per mole — and it is balanced to the last digit. That is how matter can be sustained by the vacuum without ever being a tap on it: the energy lane once read its rectenna as "tapping the wave-channel side of a flow already running," and the flow is real, but it runs both ways at once, so a static tap takes nothing (§6).

MEASURED (the QED balance) · **DERIVED** (47 nW; receipt `puthoff_zpf_balance_run`)

4. Gravity is what stored energy does to the medium

The elastic-metric law is the medium's one dynamical statement about itself: the effective length an edge presents to waves is set by the energy it carries, $\ell_{\text{eff}} = \ell_0 \cdot F(E/E_\star)$, and to first order $F = 1 + E/E_\star$, $\gamma_1 = 1$, with no free parameter. Energy *lengthens* the edge, so it gravitates. A mass is stable energy in the lattice; it lengthens the edges around it; longer edges are longer rulers, so space stretches; a clock is a standing wave on those same edges, so the clock runs slow. Light bends by the sum of the two effects — and that sum is the full general-relativistic factor of two, $4GM/c^2b$, with $\gamma_{\text{PPN}} = 1$, ray-traced to 2.000 in the network's own probe. Newton's $1/r$, gravitational redshift, and Shapiro delay follow. **DERIVED** (weak field, factor of two included — [The Shape of Gravity](#))

The consequence for how the network reasons is the whole point of this paper. **General relativity is not the source of the network's gravity; it is the weak-field shadow of the medium law.** The mainstream isomorph is Puthoff's polarizable-vacuum representation — gravity as a variable dielectric constant $K(r,t)$ of the vacuum, $\epsilon_0 \rightarrow K\epsilon_0$, $\mu_0 \rightarrow K\mu_0$, $v_L = c/K$ — which the warp paper cross-checks against the lattice law term by term ([Warp §11](#)): same sign, same bare-coupling floor, same $K < 1$ branch at strong field. "There is no perfectly rigid rod: near a mass, rulers shrink and clocks slow ... that is what 'curved space' operationally means (Dicke's and Puthoff's medium picture, and it is mainstream)" — [field guide](#). And the mechanism by which *engineered* fields curve spacetime is a theorem, not a hope: electromagnetic energy sources the metric at $8\pi G/c^4$, and Füzfa's exact Einstein–Maxwell solution for laboratory hardware (PRD 93:024014, 2016) gives the number for a superconducting coil — 1.6×10^{-25} rad per light trip, 10^{-11} only after 200 days of integration. That is the bare floor. The rest of the network is about climbing off it. **MEASURED** ($8\pi G/c^4$; Füzfa exact solution) · **OPEN** (the tensor law; the absolute scale G — §10)

PART II — THE TWO THEOREMS

5. The static theorem: a symmetric coupling to the medium is exactly zero

Three lanes of this network reached this independently, in three vocabularies, before any of them noticed the others had.

The energy lane, in Brillouin's words. A diode in thermal equilibrium cannot rectify its own Johnson noise (Brillouin 1950); the network's optical-cavity paper states this "as the correct default," names the only three escapes (source not thermal; device not in equilibrium; junction faster than the correlation time), and makes each testable. The measured controls agree: no mirror $\rightarrow$ zero output; symmetric junction $\rightarrow$ zero; "a symmetric system (no engineered asymmetry) gives exactly zero net extraction."

MEASURED

The gravity lane, in the evenness lemma. A static, or uniformly rotating, or mirror-symmetric drive on any body has $\kappa = 0$ *exactly* — "by the conservation books: it structurally cannot rectify." An incoherent Brown-class capacitor couples at the bare Einstein floor, $8\pi G/c^4 \approx 2 \times 10^{-43}$, which for a laboratory article is $\leq 3 \times 10^{-42}$ N — thirty-five to thirty-nine orders below every balance ever used. This was frozen as a pre-registration and scored: the Biefeld–Brown simulation returns $C^* = 1.58 \times 10^{-6}$ against a frozen dial of 1.4×10^{-6} , with the evenness lemma making static bootstrap "structurally impossible." PRE-REG scored • DERIVED

The root, in the vertex rule. The lattice's scattering matrix is unitary and time-reversal symmetric; "total power leaving = total power arriving" checks exactly ($\frac{1}{3}^2 + \frac{2}{3}^2 + \frac{2}{3}^2 = 1$). A closed, symmetric drive recycles; it cannot mint. DERIVED

The consequence rewrites the ledger. Every published, independently instrumented vacuum test of the Brown effect — Talley at 10^{-6} torr (1988–91), Tajmar (2004), Bartlett & Tajmar to nano-newton (2021), the comprehensive Dresden sweep of capacitors, solenoids, tunnelling currents, varistors, ϵ/μ gradients and shielded toroids at $\sim$ nN (Tajmar, Kößling & Neunzig, *Sci. Rep.* 14:19427, 2024) — tested a *static, incoherent* article. For every one of them general relativity predicts effectively zero and the medium law predicts *exactly* zero, and every one of them measured zero. Not one carried a single coherence predicate. That is the definition of NULL-STATIC: predicted zero, measured zero, a confirmation of the law and the tightest map ever drawn of which shortcuts do not exist. It is why the field guide's line "any program that starts by disputing the nulls is not doing engineering" is correct — and why the same nulls are not evidence against the medium: they never reached it. Montgolfier's own 1959 torsion balance at 5×10^{-5} mm Hg put Brown's disc–wire article at *nothing* under DC. As predicted.

6. The transient theorem: an asymmetric, time-broken coupling is not zero

Energy: measured where the boundary is driven. The result is exact: the dynamical Casimir effect. Change a boundary in time and the vacuum yields real quanta, with production set by how *abruptly* the boundary changes (Wilson 2011; the count diverges only for infinitely fast removal). Static boundary: nothing. Symmetric oscillation: photons in pairs, no net. Abrupt, one-sided: real output — paid for by whatever moves the boundary. **MEASURED**

Energy: open where the boundary is static. The network's best-known energy anchor is a different kind of device, and draft 0.1 of this paper filed it in the wrong column. Model's devices (*Symmetry* 13:517, 2021) — a metal–insulator–metal junction with a one-sided mirror at a nanometre gap — report 1.4 pW per $0.02 \mu\text{m}^2$ at 33 nm — 70 W/m² — across a thousand-plus devices in 21 batches with eight or nine artefact classes excluded, a clean 1/d law from 33 to 1100 nm, and no output without the mirror. **MEASURED** (anchor; no independent replication yet). But nothing in that device moves: it is a static geometry. A femtosecond junction is fast, not driven — the fluctuation–dissipation theorem holds at every frequency, so speed buys bandwidth, not permission — and a one-sided mirror changes how fast the junction trades with the field on that side, not the level it trades at. At one temperature a static device's forward and reverse flows cancel term by term, asymmetric or not (The Zero-Point Machine §8; Model & Dmitriyeva reached the same verdict on rectifying the zero-point field in 2019). So if the current is real, it draws on something the device description does not name — a residual temperature gradient, light, RF, or stored charge — or it is the first static violation of the fluctuation–dissipation theorem, which would be a bigger discovery than any energy source. One run decides: the device sealed in a dark, verified-isothermal metal box beside its no-mirror and reversed twins, with the delivered energy integrated past its own stored-energy bound. **OPEN** (the source) · **PRE-REG** (proposed: the sealed-isothermal run)

Force: open, and every positive in the record is transient. The same law says the only regime in which a body can rectify the medium into a force is a time-asymmetric, geometrically asymmetric drive — $\kappa \neq 0$. And when the electrogravitics lineage is read in its own documents rather than its legend, that is the only regime in which anyone ever saw anything in vacuum:

Source	Drive	Vacuum result	Register
Brown, Notebook §§42–43, Apr 1956 — "The Impulse Effect"	vacuum spark	force "concurrent with the vacuum spark," with the recovery of potential — not with the static field; an external chopper (voltage change without spark) gave nothing	MEASURED (his own hand)
Montgolfier 1957 report, bell jar 2–5×10 ⁻⁵ mm Hg	flash-driven	continuous rotation of the articles, polarity-locked negative→positive, killed by a grounded screen; unmeasurable (bell braking); the control turnstile that spins at 40 kV in air does not turn at 150 kV in vacuum	MEASURED (sponsor's own report, read in the original)
Montgolfier 1959 final report, 1.48 m chamber, torsion balance, DC ±50–55 kV	static DC	Brown disc–wire: <i>nothing</i> ; two other articles: nothing; plexiglas capacitor: 3–4°; BaTiO ₃ -loaded capacitor: 30° ≈ 1 g·cm, reversing with polarity — no current, breakdown, or lever-arm log survives	NULL-STATIC (Brown geometry) · OPEN (BaTiO ₃ datum — see the loose end below)
Bahnson notebooks, May 1958	AC vs biased	"AC: none" on the tent rig; small only when biased — symmetric AC on an asymmetric capacitor is <i>predicted</i> null	NULL-STATIC
Talley, AFAL/PL 1988–91, 10 ⁻⁶ torr	DC vs breakdown / pulsed PZT	no DC thrust; forces only during breakdown and with pulsed piezo; his own unfollowed recommendation: high-K, pulsed	NULL-STATIC (DC) · OPEN (pulsed)
Buhler / Exodus, 2023–26	stacked dielectric, DC-persistent	5–10 mN claimed across ~2000 vacuum runs; unpublished, unreplicated; claimed force fell ~40× under scrutiny	OPEN (declining under scrutiny)

So Brown *did* see a positive in France, in vacuum, in the transient regime — the 1957 flash rotations — and it is exactly the regime the law allows. It is also exactly the regime where the conventional explanation lives, and the two have never been separated. A vacuum flash is a cathodic arc; cathodic arcs eject metal plasma at ~10⁴ m/s, ~10⁻⁵ N·s per coulomb, *always toward the positive electrode*, killed by a grounded screen, present only during the spark. That is point-for-point the 1957 phenomenology,

and Cornillon proposed it himself ("anodic effect"). Brown's chopper-vs-spark note cuts the same way — or, in the energy lane's reading, is an *edge-rate* datum: a spark's edge is a thousand times faster than a 1956 chopper's, and DCE production scales with abruptness. Four discriminators would settle it and no one has ever run them: impulse per coulomb (arc recoil is pinned at $\sim 10^{-5}$ N·s/C; a medium coupling is not); electrode mass loss (an arc is eroded metal); chopper-vs-spark at fixed charge and edge rate; and force per coulomb against dielectric K — the one lever ejecta cannot cover. That is bench item 6 of the field guide, and it is a weekend on a torsion balance. [OPEN](#) — the transient channel, undiscriminated

The loose end, stated against ourselves. The 1959 BaTiO₃ datum (30°, ≈ 1 g·cm, reversing with polarity) was on the DC runs. A static positive is what the evenness lemma says cannot exist. Either it is an artefact or an unlogged breakdown — no current or flash log survives, which is the record problem of The Missing Record — or high-K dielectrics do something the static theorem does not cover. It is the one number in the lineage that mildly cuts against the medium law, and it is why K-scaling is on the bench list rather than in a footnote.

7. One machine, three readings

The energy lane and the gravity lane were written by different sessions for different readers, and each has been treating the other's device as a neighbour. They are the same device. The network's shell note says it in one sentence:

"A driven, asymmetric current/magnetic pattern is an open sub-circuit whose 'return path' must close through something beyond the craft — exactly the structure Model's device has electrically ... One asymmetric drive, three readings: power out, force out, weight change — depending on what the return path couples to." — *The Shell Is the Reactor* §2

Line the two up and every element corresponds:

Element	Energy reading (measured when driven)	Force reading (open)
Boundary asymmetry	a one-sided boundary driven in time (dynamical Casimir effect); a static mirror changes rates, not levels	asymmetric hull / capacitor geometry; sets the <i>direction</i> of κ

Rectifying asymmetry	work-function step $\Delta\phi$ — needed, not sufficient: a real gradient or drive must also be present; symmetric junction $\rightarrow$ zero	time-asymmetric drive cycle $\kappa \neq 0$; static or symmetric AC $\rightarrow$ zero
Speed	the drive's edge rate (DCE $\propto$ abruptness); a fast junction sets bandwidth, not permission	spark edge (ns) vs chopper (μ s–ms); DCE $\propto$ abruptness
What is drawn on	the driver's work, or a real temperature gradient; the zero-point floor is matched, not cold	the loaded edges at $E_\star$; the lattice is the momentum sink
Signature	a static device nulls in a sealed isothermal box; a gradient engine cools its hot side and names its cold body	hover is a <i>lock</i> at $\approx$ zero power; sign reverses under drive flip; chamber-independent
Books	conversion, not creation: the driver or the gradient pays; conservation on the joint ledger	propellantless, not reactionless: the medium carries the reaction; conservation on the joint ledger
Return path closes through	the rest of the circuit $\rightarrow$ power	the medium $\rightarrow$ force; the medium under gravity $\rightarrow$ weight change

This is why the field guide's §1.2 could *import* the energy lane to derive that only transients work, and why the shell note's third reading — weight change — is the electrogravitics claim in energy-lane clothing. The prediction it makes is falsifiable in both directions: a device that produces power should, when its return path is closed through the medium instead of the circuit, produce force of the same asymmetry; and a device that produces force must pay for it from its drive, visibly, in its own power books. [OPEN](#) — a hypothesis, named as such in its source

PART III — THE BOOKS

8. Conservation is kept by the medium, not violated

The reason "antigravity" and "free energy" are filed with the frauds is that their loudest advocates claim conservation violations. The medium picture never does; it relocates the ledger.

Propellantless is not reactionless. A reactionless drive — a force with no reaction — is forbidden by Noether's theorem and stays forbidden here. A propellantless drive carries no reaction *mass*; the reaction momentum is real and is carried by a field or medium — photons for a solar sail, the rails for a maglev, and, here, the lattice. "The single honest test of any propellantless claim is: where does the reaction momentum go?" — [Warp](#) §2. Puthoff & Little (2002 §2.1, App. A) state the identical constraint: static fields cannot self-propel because of hidden momentum; a propellantless drive must couple to the external metric. The medium law answers "the lattice," and the crux of the self-propulsion claim is the lattice-mode speed v_L : if $v_L \approx c$ the drive is a photon rocket ($\sim nN/W$); only a slow, dense momentum-carrying lattice mode ($v_L \ll c$) gives liftable thrust. That is "the single dynamical assumption a saucer-scale claim rests on. State it; do not smuggle it in." [DERIVED](#) (books) · [OPEN](#) (v_L)

Extraction is not free energy. The first law is the most-tested statement in physics and "net extraction is lawful in exactly one way: draining a real reservoir (an open system)." The vertex rule is unitary — the lattice cannot mint. The vacuum edges are loaded to $E_\star$ — but a loading at the ground level is a floor, not a reservoir: a ground state is *passive*, and no cyclic process run on it returns more work than it puts in (Pusz & Woronowicz 1978 — the quantum form of "a level, not a stockpile"). An open system therefore needs a reservoir that is not at its floor: a real temperature gradient (a heat pump's $COP > 1$ is heat moved from a colder body, not energy from nowhere), light, or a driver whose work the vacuum converts (the dynamical Casimir effect). The closed-system rule stays intact: a closed vortex cannot recycle above 100%; a shaft between a motor and its own generator is inside the inventory, not outside it, and spins down under load every time. The network's phrase: reclassify such claims from "wrong physics (forbidden)" to "wrong *words* for a lawful open-system coupling," and then ask the only honest question — which reservoir, not at its floor, is paying? — which is a bench number: unproven for gradients and drives, excluded for the static floor. [DERIVED](#) (bookkeeping; passivity) · [OPEN](#) (net extraction from gradients and drives)

9. The dial: coherence × asymmetry × consonance × wholeness

The force law of the network is $F = -\nabla[C_{\text{sub}} \cdot |\chi_v|^2 \cdot U_{\text{drive}}]$: the gradient of stored coherent energy, times how well the substrate's own coherent mode overlaps the lattice's ($|\chi_v|^2$), times what fraction of the overlapping energy the substrate converts

into force (C_{sub}). "It scales with stored coherent energy U_{drive} , not dissipated power ... Higher $Q \rightarrow$ more stored energy $\rightarrow$ larger force, at fixed input power." Every piece is derived or controllable except C_{sub} and the directional derivative — the one open number, and the bench's to decide. **DERIVED** (form) · **OPEN** (magnitude)

The scale of the climb is the whole story of the shelf. An incoherent lump — Brown's capacitor, every published null article — sits at the bare-vacuum floor of C , the Einstein constant $8\pi G/c^4 \approx 2 \times 10^{-43}$. Hover needs $C \approx 10^{-6}$. The consonance ceiling is $e^{(-P/12)} \approx 0.873$. The tightest published null bounds static capacitors at $C < 2 \times 10^{-10}$ — thirty-three orders above the floor, four below Buhler's implied value. Thirty-seven orders, on one dial, and the four predicates that move it are the four the warp paper reads as Steiner's four ethers, now the four factors of the thrust: *coherence* (a phase-locked condensate boundary — Cooper pairs, a bulk-acoustic plate mode, an optical cavity mode; thermal motion averages to nothing); *asymmetry* ($\kappa \neq 0$ — a handed inward spiral is the natural form, because the lattice's own loops are chiral: pentagons force $\cos 36^\circ = \phi/2$); *consonance* (icosahedral / ϕ -ratio geometry, degree-3 carbon cages, quasicrystals — a substrate whose symmetry is the lattice's own accumulates across its volume instead of cancelling); *wholeness* (how much of the body moves as one). Not one published test article carried even one of these; the first null that carries all four, instrumented and verified present, will be the first null that reaches the medium law. "A target, not a rescue." **DERIVED** (predicates) · **OPEN** (where a real body sits on the dial)

Two regimes fall out of the same law. **Hover is statics**: engage the coupling and the lattice pins the pose at $\approx$ zero power, the way a rail pins a maglev — "lock, not lift"; a persistent force on a balance is the support mode, the medium holding the device and doing no work, which is why "a charged object accelerates forever in vacuum" would be the reactionless red flag and the support reading is the conservation-honest one. **Travel is dynamics**: acceleration costs work, recoverable regenerative-braking style. Driven to saturation in a closed shell the same law is the $K < 1$ branch of the charged-mass solution — the Alcubierre / Morris–Thorne class, GR-legal, gated on driving the perturbation to $O(1)$. Only flat-space FTL is forbidden. **DERIVED** (regimes) · gated

10. What the medium still owes

A first-principles paper that hid its open seams would be a pitch. These are the medium's, in order of weight:

1. **Lorentz invariance of the medium.** Half paid. The loading has a known frame-free form: $\hbar\omega/2$ per mode — an energy spectrum rising as ω^3 — is the one loading that looks identical from every inertial frame, because the Doppler shifts of a boost cancel exactly for ω^3 and for no other power (Marshall 1965; Boyer 1969; receipt `puthoff_zpf_balance_run`, P9). A thermal loading has a wind — the cosmic background's 3.36 mK dipole, predicted from our 370 km/s and measured, is exactly one — and the zero-point loading does not. **DERIVED** (the loading). What is still owed is the lattice. The scalar law reproduces every weak-field ratio of general relativity, but the vertex rule is built from unitarity + threefold symmetry + time reversal — none of which is a boost — and the G-vertex derivation speaks of "one vertex per Planck-scale location." "One vertex per Planck location" and "every edge loaded to $E_\star$ " must be read as the *cutoff* of an ω^3 loading, not as a level per edge in a preferred frame: linear Lorentz violation is already excluded beyond ten Planck energies (LHAASO, GRB 221009A). **OPEN** — the first seam, now the lattice's dispersion rather than the loading.
2. **Inertia.** No paper in the root set addresses it. The one mainstream attempt to make inertia a vacuum effect (Haisch–Rueda–Puthoff) is refuted on its own terms (Levin, PRA 79:012114, 2009 — in the network's gravity deep-dive), and the network does not lean on it. In a medium picture inertia should be the medium's resistance to a pattern's change of state — the obvious conjecture, and unwritten. **OPEN** — the second seam.
3. **Magnitude.** "The chain does not derive Newton's constant. Nothing else was ever missing." $E_\star$ is either Planck-by-definition (and the lepton masses measure G) or the masses fix $E_\star$; the network has proven the current axioms do not decide it (object #8, the κ -shift countermodel). The absolute scale of the loaded medium is a metrology question, not yet a derivation. **OPEN**
4. **The tensor law.** Scalar F closes the weak field; the nonlinear, steerable-bubble regime needs the full equation. **OPEN**
5. **v_L , the lattice-mode speed** — §8. **OPEN**

And what is *closed*, so that no reader mistakes this paper for the literature it is often shelved beside: the stored Casimir energy as fuel ($12 \mu\text{J}/\text{m}^2$, exhausted in 172 ns, "eight orders short"); Schwinger pair mining ("12.6 orders short"); Unruh / zero-point-field propulsion (the bath's coherence time $\tau = 2\pi c/a$ is set by the very acceleration that makes it — "the vacuum tears before the bath pays for itself"); the "vast vacuum energy" as a resource (the 121-order error); the Haisch–Rueda–Puthoff inertia (Levin 2009); a self-running closed vortex or motor–generator loop (inside its own inventory); the zero-point floor as a cold sink, and static rectification of it (it cancels from every net flow — The Zero-Point Machine §8). Each is priced in the network's own papers with the claim taken at its most favourable. **REFUTED** — do not re-open

11. Why the network wrote "known physics only" — and what it actually believes

The field guide opens "This paper makes no new physics claim." The energy partner brief says "none of which requires new physics." Read next to §§1–9 those sentences look like a contradiction. They are a *register*, chosen on a particular day for a particular reader — a drone company that would stop reading at the first sentence it could dismiss — and the choice was correct for that reader. The mechanism by which fields curve spacetime is a theorem; the static nulls are binding; the bench program is the same whichever ontology you hold. So those papers were written from the floor up.

But the floor is not the network's belief, and it is time to say so on the front page. The network's belief is §§1–9: the medium. It is what every paper behind this one reasons *from* once past its opening — "the lattice carries the reaction," "energy lengthens the edges," "the lattice pins the pose," "Dicke's and Puthoff's medium picture." "Known physics" is the network's *floor*, not its ceiling: nothing here contradicts a measurement, and where the medium law and general relativity both speak they agree to the last decimal the weak field can test. Where they part — the transient coherent channel, the loaded vacuum under a driven boundary, $K < 1$ — the medium makes predictions and general relativity is silent, and the network's bench is built to test exactly those. That is not less rigorous than "known physics only." It is the same rigour with its premises stated.

12. The record

One more thing has to be on the front page, because it changes how the null ledger reads. In this field, "no published positive" and "no positive" are not the same sentence. Brown's Notebook Vol. 3 was "not released"; his 1975 paper's "publication was withheld"; his 1959 GE King-of-Prussia vacuum notes "were never published and are not available to me now" (his words, 1973). Montgolfier's torsion runs carry no surviving current or breakdown log. Tesla's effects were taken into government custody within days of his death. The United States has had, since 1951, a statute for classifying private inventions, with some six thousand secrecy orders standing in any given year. Schauberger signed away "all of his implosion work — past, present, and future" twelve days before he died. The companion paper The Missing Record lays that out — mechanism, cases, and the counter-cases where the file points the other way (Keely; Carr) — with the register of each. Its consequence for this paper is one clause: an absent receipt in this field is as often a withheld one as an absent effect, and that is *why* the network runs its own bench rather than reading the literature's silence as an answer. It licenses a bench. It licenses no claim.

13. The chain, on one page

#	Link	Where it lives	Register
1	Space is a medium — a loaded elastic network; screen = 12-pentagon Goldberg tiling; $\Lambda = 3/N$	Shape §1–2, §5.1, §9	DERIVED / OPEN (graph)
2	The medium is loaded — every vacuum edge at $E\star = 2m_{\text{edge}} c^2$; Lamb, Casimir, DCE are the measured twins; a level, not a stockpile; $\hbar\omega/2$ per mode — the one frame-free loading	Shape §8, §11.1; energy lane; §10	DERIVED / MEASURED
3	Matter is a pattern the medium sustains — at the atom, a balance: ≈ 47 nW in and out per hydrogen atom, cancelling exactly (Puthoff 1987; QED); the electron as light closed on itself, $\alpha/2\pi$ leakage as its field	§3; The Electron §1–6	MEASURED (atom) / OPEN (electron, conditional)
4	Gravity is what stored energy does to the medium — $F = 1 + E/E\star$, $\gamma_1 = 1$, factor of two; GR is the	Shape of Gravity; Warp §11	DERIVED (scalar) / OPEN (tensor)

	shadow; PV is the mainstream isomorph		
5	Static theorem — static coupling at one temperature is exactly zero, symmetric or not (Brillouin = evenness lemma = unitary vertex = fluctuation–dissipation); a cavity changes rates, not levels; every published null is predicted	Optical-cavity §1d; field guide §1.1, §4; BB pre-reg	DERIVED / NULL-STATIC
6	Transient theorem — driven, time-broken coupling is not zero: light (dynamical Casimir effect, driver-paid) / force (every vacuum positive is transient; undiscriminated from arc recoil). The static 70 W/m ² current is a static device: source open, sealed-isothermal run	§6; field guide §1.2; Brown, Montgolfier, Talley	MEASURED (DCE) / OPEN (static current, force)
7	One machine, three readings — power / force / weight by return path	Shell Is the Reactor §2	OPEN (hypothesis)
8	Conservation kept by the medium — propellantless ≠ reactionless; extraction ≠ free energy; the zero-point floor is passive, so the payer is a gradient or a driver; the closed-system rule intact	Warp §2; energy optimum §2	DERIVED
9	The dial — $F = -\nabla[C xv ^2U]$; floor $2 \times 10^{-43} \rightarrow$ hover $10^{-6} \rightarrow$ ceiling 0.873; four predicates; hover = lock	xv force law; Warp §4, §13; field guide §4	DERIVED (form) / OPEN (magnitude)
10	Owed — the lattice's dispersion (the ω^3 loading pays the spectral half of Lorentz); inertia; $G / E\star$; the tensor law; v_L	§10	OPEN
11	The record — withheld ≠ absent; licenses a bench, not a claim	The Missing Record	documented / open, per case

Sources

In-network: The Shape of the Universe (§1–2 edges as the frame; §4 vertex rule; §5.1 Goldberg screen; §8 matter as settled edge-energy; §9 $\Lambda = 3/N$; §11.1 elastic-metric law) · [The Shape of Gravity](#) · The Electron §1–6 · [Electrogravitics](#) §1.1–1.2, §3, §4, §6 · [The Warp Drive](#) §2, §4, §8, §11, §13 · The Optical-Cavity Device §0b, §1c, §1d, §5 · The Zero-Point Machine §7–§8 · *PUTHOFF_1987_ZPF_BALANCE_OPH_OVERLAY* (2026-09-23; receipt `puthoff_zpf_balance_run`, 9/9) · Casimir Energy Extraction §0–3 · Energy Generation Optimum §2 · [Townsend Brown](#) §3, §7 · *The Shell Is the Reactor* (2026-07-20) · *CHI_NU_FORCE_LAW* · *G_VERTEX_DERIVATION* ·

ELASTIC_METRIC_BACKREACTION_LAW · MONTGOLFIER_REPORT_RETRIEVAL (2026-08-18) · RESULTS_BIEFELD_BROWN (2026-08-13) · GRAVITY_ENGINEERING_DEEPDIVE (2026-08-13) · ISF/Haramain audit (2026-08-14).

External: Brillouin, L., "Can the rectifier become a thermodynamical demon?" *Phys. Rev.* 78, 627 (1950) · Moddel, G. et al., *Symmetry* 13, 517 (2021); Moddel & Dmitriyeva, "Extraction of zero-point energy from the vacuum," *Atoms* 7, 51 (2019) · Wilson, C. M. et al., "Observation of the dynamical Casimir effect in a superconducting circuit," *Nature* 479, 376 (2011) · Puthoff, H. E., "Ground state of hydrogen as a zero-point-fluctuation-determined state," *Phys. Rev. D* 35, 3266 (1987) · Dalibard, J., Dupont-Roc, J. & Cohen-Tannoudji, C., "Vacuum fluctuations and radiation reaction: identification of their respective contributions," *J. Physique* 43, 1617 (1982) · Cetto, A. M. & de la Peña, L., *Phys. Rev. A* 37, 1952 (1988) · Marshall, T. W., *Nuovo Cimento* 38, 206 (1965); Boyer, T. H., *Phys. Rev.* 182, 1374 (1969) · Pusz, W. & Woronowicz, S. L., "Passive states and KMS states for general quantum systems," *Commun. Math. Phys.* 58, 273 (1978) · Assawaworrarit, S., Omair, Z. & Fan, S., *Appl. Phys. Lett.* 120, 143901 (2022) · Cao, Z. et al. (LHAASO), *Phys. Rev. Lett.* 133, 071501 (2024) · Lamoreaux, S. K., *PRL* 78, 5 (1997); Mohideen & Roy, *PRL* 81, 4549 (1998) · Dicke, R. H., "Gravitation without a principle of equivalence," *Rev. Mod. Phys.* 29, 363 (1957) · Volkov, Izmet'sev & Skrotskii, *Sov. Phys. JETP* 32, 686 (1971) · Sakharov, A. D., *Sov. Phys. Dokl.* 12, 1040 (1968) · Puthoff, H. E., "Polarizable-vacuum (PV) representation of general relativity," gr-qc/9909037; Puthoff, Little & Ibison, "Engineering the zero-point field and polarizable vacuum for interstellar flight," *JBIS* 55, 137 (2002) · Füzfa, A., "How current loops and solenoids curve spacetime," *Phys. Rev. D* 93, 024014 (2016) · Levin, Y., "Inertia as a zero-point-field force: critical analysis of the Haisch–Rueda–Puthoff inertia theory," *Phys. Rev. A* 79, 012114 (2009) · Talley, R. L., "Twenty-first century propulsion concept," PL-TR-91-3009 (1991) · Tajmar, M. & Schreiber, T., *J. Electrostatics* 107, 103477 (2020); Bartlett & Tajmar, *Acta Astronautica* 181, 12 (2021); Tajmar, Kößling & Neunzig, *Sci. Rep.* 14, 19427 (2024) · Cady, W. M., ONR Pasadena File 24-185 (15 Sep 1952) · Sud Aviation, "Projet Montgolfier" (1 Jul 1957) and *Rapport final* (Courbevoie, 15 Apr 1959) · Brown, T. T., Notebooks §§42–43 (Apr 1956), §67 · Bahnsen Laboratories notebooks (1957–59) · Anders, S. & Anders, A., "Emission of neutral atoms and macroparticles from vacuum arcs" / vacuum-arc thruster literature ($\sim 10\text{--}20\ \mu\text{N/A}$).

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